

IDBI DEFAULT PREDICTION

Step 1: What data do we need?

Think of a customer like a patient. To diagnose health, a doctor checks blood pressure, heart rate, medical history, lifestyle, and symptoms. Similarly, to predict loan default, we need multiple views of a customer's financial health.

We can divide the data into **6 major categories**.

1. Customer Profile

This tells us **who the borrower is**.

Feature	Why it matters
Customer ID	Unique identifier
Age	Different age groups have different risk profiles
Gender	Optional (may be excluded to avoid bias)
Marital Status	Family responsibilities affect repayment
Education Level	Higher education often correlates with stable income
Occupation	Government job vs freelancer vs business
Employment Type	Salaried, self-employed, unemployed
Years at Current Job	Stable employment generally reduces risk
Annual Income	Ability to repay
Number of Dependents	Higher expenses can increase financial stress

Residential Status Owned house or rented

Example:

Customer

Age: 27

Occupation: IT Engineer

Salary: ₹11 LPA

Employment: Salaried

Years in Job: 5

2. Loan Information

This describes the loan itself.

Feature	Why it matters
Loan Amount	Larger loans may carry higher risk
Loan Type	Home, Personal, Vehicle, MSME
Interest Rate	High rates can make repayment harder
Loan Tenure	Longer loans mean longer exposure
EMI Amount	Monthly repayment obligation
Remaining Loan Balance	Outstanding debt
Loan-to-Income Ratio	Measures affordability

Example

Loan Amount

₹12,00,000

EMI

₹26,000

Salary

₹45,000

EMI consumes 58% of salary

High Risk

3. Repayment History (Most Important)

Past behaviour is one of the strongest predictors of future behaviour.

Feature	Why it matters
Number of EMIs paid on time	Positive repayment history
Number of delayed payments	Indicates repayment issues
Maximum days overdue	Severity of delay
Consecutive missed EMIs	Strong default indicator
Last payment date	Shows payment recency
Partial payments	Financial stress signal

Example

EMI History

Jan ✓

Feb ✓

Mar Late

Apr Late

May Missed

Jun Missed

The model immediately identifies this as high risk.

4. Transaction Behaviour

Banks learn a lot from how money moves.

Feature	Why it matters
Monthly Income Credits	Stable income is reassuring
Average Monthly Balance	Financial cushion
Number of Transactions	Financial activity level
Cash Withdrawals	Large increases may indicate distress
Utility Bill Payments	Missing bills may signal cash flow issues
Salary Regularity	Delayed or missing salary credits
Spending Categories	Lifestyle changes

Example

Previous Salary

₹80,000

Current Salary

₹52,000

Income dropped

35%

Risk ↑

5. Credit Behaviour

This reflects borrowing habits.

Feature	Why it matters
Credit Score (e.g., CIBIL)	Overall creditworthiness
Number of Active Loans	Existing debt burden
Credit Card Utilization	High usage suggests financial pressure
Recent Loan Applications	Multiple applications can indicate liquidity needs
Credit History Length	Longer history provides more confidence

Example

Credit Score

820 → 760 → 690

Risk increasing

6. Alternative Data (Innovative)

Traditional data doesn't tell the full story. Alternative data helps identify risk earlier.

Feature	Why it matters
GST Returns	Business performance for MSMEs
UPI Transactions	Spending and income patterns
EPFO Contributions	Employment continuity

Account Aggregator Data Consolidated financial view

Electricity Bills Regular payment behaviour

Mobile Recharge
Frequency Lifestyle consistency

Step 2: Feature Engineering

Raw data isn't enough. We derive new metrics that reveal hidden patterns.

Instead of:

Salary

₹75,000

We calculate:

Salary Growth

+8%

Income Stability

95%

Income Volatility

Low

These engineered features are often more predictive.

Income Stability Score

Based on salary consistency.

Month

Jan ₹50k

Feb ₹50k

Mar ₹49k

Apr ₹50k

May ₹51k

Stable

Compare with:

Jan ₹80k

Feb ₹75k

Mar ₹45k

Apr ₹35k

May ₹22k

Very unstable

EMI Burden Ratio

One of the strongest indicators.

Formula:

$\text{EMI Burden} = \text{EMI} / \text{Monthly Income}$

Example:

Income

₹50,000

EMI

₹32,000

EMI Burden

64%

Anything above ~50% deserves attention.

Credit Utilization

Credit Card Limit

₹2,00,000

Spent

₹1,80,000

Utilization

90%

High utilization often correlates with financial stress.

Missed Payment Ratio

Paid EMIs

20

Missed

5

Ratio

20%

Spending Trend

Jan ₹30k

Feb ₹31k

Mar ₹29k

Apr ₹72k

May ₹85k

A sudden spike may indicate unusual financial behaviour.

Savings Trend

Average Balance

Jan ₹1.5L

Feb ₹1.3L

Mar ₹1.0L

Apr ₹72k

May ₹35k

Rapidly declining savings can be an early warning.

Salary Trend

Jan ₹70k

Feb ₹70k

Mar ₹70k

Apr ₹58k

May ₹45k

This is a strong predictor of repayment risk.

Step 3: Final Dataset

Each row represents one customer, and the model learns from these features.

Custo mer	Ag e	Inco me	Loan Amou nt	EMI Burd en	Cred it Scor e	Miss ed EMIs	Salary Trend	Saving s Trend	Credit Utilizati on	Defa ult
C001	29	65,000	5L	28%	790	0	Stable	Stable	32%	No
C002	41	45,000	12L	62%	640	4	Declini ng	Declini ng	91%	Yes
C003	35	80,000	8L	34%	760	1	Stable	Stable	45%	No

The **Default** column is the target variable the model learns to predict.

DATASET

public loan dataset (such as Home Credit Default Risk or LendingClub) and enrich it with **simulated transaction history**. For example, generate six months of salary credits, EMI payments, account balances, and credit utilization trends for each customer. This lets you demonstrate time-series analysis, feature engineering, and explainable AI without needing confidential banking records.

This approach is realistic, technically impressive, and practical within your 5-day timeline. It also gives you a much richer demo than training a model on a static CSV alone.

1. Demographic Features

Feature	Why?
Age	Younger borrowers may have unstable careers
Education Level	Financial stability indicator
City/Tier	Economic conditions differ by region
Family Size	More dependents = more expenses
Residential Status	Own/Rent/Company accommodation
Years at Current Address	Frequent moves may indicate instability

2. Employment Features

Instead of just salary.

Feature	Description
Current Salary	
Salary Growth (%)	
Number of Job Changes	
Years with Current Employer	

Employer Category (Govt/Private/Startup)

Employment Stability Score

Promotion Frequency

Industry Type

Example

Salary Trend

Jan 60k

Feb 60k

Mar 58k

Apr 52k

May 48k

Jun 45k

Instead of using six values, calculate

- Salary Growth
- Salary Variance
- Salary Stability Score

3. Banking Behaviour Features

These are extremely useful.

Average Monthly Balance

Someone whose account constantly stays below ₹5000 is more risky.

Minimum Monthly Balance

One bad month might be acceptable.

Several bad months indicate stress.

Balance Volatility

If balance changes wildly every month

2L

30k

1.8L

20k

1.7L

15k

That customer has unstable cash flow.

Number of Overdrafts

Customer frequently exceeds account balance.

Huge red flag.

Returned Cheques

NSF (Insufficient Funds)

Banks love this feature.

4. Transaction Behaviour

Instead of raw transactions.

Calculate

Monthly Spending

Jan 28k

Feb 29k

Mar 30k

Apr 68k

May 74k

Feature

Spending Increase %

Savings Rate

Formula

Savings

=

Income

-

Expenses

Low savings indicate less ability to handle emergencies.

Income Regularity

Salary arrives

1st

1st

1st

15th

No Salary

Huge warning.

Cash Withdrawal Ratio

Customer withdrawing almost everything in cash

↓

Possible financial stress.

Digital Payment Ratio

UPI

Card

Cash

Net Banking

Different behaviour patterns can indicate changing habits.

5. Loan Features

Instead of Loan Amount

Create

Loan-to-Income Ratio

Loan

Income

Example

Loan

20 Lakh

Income

4 Lakh

Ratio = 5

Very risky.

EMI-to-Income Ratio

One of the strongest predictors.

EMI

Salary

Remaining Loan Percentage

Outstanding Loan

Original Loan

Customer has paid

90%

↓

Very different risk compared to

Paid

10%

Multiple Loans

Count

Personal Loan

Home Loan

Vehicle Loan

Gold Loan

Credit Card

More loans

↓

Higher risk.

6. Credit Bureau Features

These are extremely important.

Feature

CIBIL Score

Credit Score Trend

Number of Credit Cards

Credit Utilization

Number of Hard Inquiries

Number of Closed Loans

Total Outstanding Debt

7. Behavioural Features

These are often missing in student projects.

Loan Prepayments

Customers paying early

↓

Low risk.

EMI Payment Timing

Customer pays

1 day before due date

Every month

↓

Excellent.

Another customer pays

10 days late

Every month

↓

Risk.

Customer Complaints

Lots of complaints

↓

Potential financial issues.

Customer Service Calls

Calls asking

"Can I postpone EMI?"

Very useful.

Login Frequency

If using bank app.

Sudden inactivity

↓

Possible concern.

8. Time-Series Features

Instead of using only today's values.

Use history.

Example

Last 6 Months

Credit Score

780

770

760

730

700

670

Create

Credit Score Trend

↓

Declining.

Same for

Salary

Savings

Balance

Credit Utilization

EMI

UPI

GST

Everything.

Banks care more about **direction** than **current value**.

9. Macroeconomic Features

Very few teams think of these.

Examples

Interest Rate

Inflation

Fuel Price

Industry Layoffs

Regional Unemployment

Example

IT layoffs

↓

Software Engineers

↓

Higher default probability.

10. AI Extracted Features (LLM)

Suppose customer email

I lost my job.

Can I delay EMI?

LLM extracts

Job Loss = True

Financial Distress = High

Urgency = High

Sentiment = Negative

These become ML features.

Explainable AI Features

Instead of simply showing

Probability

89%

Show

Reason

↓

Salary decreased 28%

↓

Credit utilization 91%

↓

3 delayed EMIs

↓

Loan-to-income ratio high

↓

Savings declining

↓

Credit score dropped 95 points

Judges LOVE this.

Dataset Advice

Option 1 (Best)

Home Credit Default Risk (Kaggle)

Pros

- 300,000+ customers
- Rich feature set (100+ columns)
- Realistic banking data
- Widely used in industry
- Ideal for explainability

Cons

- Very large (needs preprocessing)
-

Option 2

LendingClub Loan Dataset

Pros

- Loan-centric
 - Includes repayment status
 - Easier to understand
 - Good for binary classification
-

Option 3

Give Me Some Credit (Kaggle)

Pros

- Clean
- Small
- Easy to train

Cons

- Fewer features
-

Option 4

German Credit Dataset (UCI)

Good for demonstrations.

Not ideal for a national hackathon.

My Recommendation

I would **combine datasets** rather than relying on a single one:

- **Base dataset:** Home Credit Default Risk (rich customer and loan features)
- **Supplementary dataset:** LendingClub (loan repayment behavior)
- **Synthetic data generation:** Create 6-12 months of transaction history for each customer (salary credits, balances, EMI payments, spending trends, UPI usage) using realistic rules.

This hybrid approach lets you showcase static credit data, behavioral trends, and time-series analysis, which is much closer to what banks actually do.

One "Wow" Feature

Create a **Financial Health Score (0-100)** from your engineered features.

Example:

Customer Financial Health Score

84 / 100

Income Stability 90%

Repayment Behaviour 85%

Credit Utilization 60%

Savings Trend 80%

Debt Burden 50%

Overall Risk

LOW

Proposed Methodology

Phase 1: Enterprise Data Integration

Banks have data scattered across dozens of systems.

Instead of relying only on CBS (Core Banking System), the platform ingests data from multiple sources.

Internal Sources

- Core Banking System
- Loan Management System
- Credit Card System
- Internet Banking
- Mobile Banking
- UPI Transactions
- CRM
- Collection Department
- Call Centre
- Branch Visits
- Complaint Portal

External Sources

- CIBIL / Experian
- GST
- Income Tax (where permitted)
- EPFO
- Account Aggregator
- RBI Public Data
- Economic Indicators
- Bureau Reports

Instead of seeing only

Loan = ₹10 Lakhs

the system understands

This customer recently changed jobs, salary has reduced, GST filing is delayed, credit utilization increased, and savings have declined.

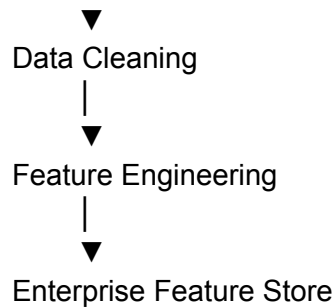
This creates a **360° Financial Profile**.

Phase 2: Data Lake & Feature Store

Instead of directly training on raw data, build a centralized feature repository.

Raw Banking Data

|



The feature store ensures consistency across:

- Loan approval
- Credit monitoring
- Fraud detection
- Marketing
- Portfolio analytics

This is how banks scale AI.

Phase 3: Dynamic Financial Health Engine

Rather than calculating only a **Probability of Default**, calculate multiple business scores.

Credit Health Score

Ability to repay.

Liquidity Score

Cash flow available.

Behaviour Score

Repayment discipline.

Stability Score

Income consistency.

Debt Burden Score

Debt relative to income.

Financial Stress Index

Combines

- Salary decline
- Credit utilization
- Missed EMI
- Savings reduction
- Spending volatility

Instead of

Default Probability

74%

The dashboard shows

Financial Health Score

61/100

Liquidity

Moderate

Behaviour

Excellent

Debt Burden

High

Stress Level

Increasing

Business users understand this immediately.

Phase 4: Explainable AI Risk Engine

Prediction alone doesn't satisfy regulators.

For every prediction, provide reasons.

Example

Customer

Default Risk

82%

Contributing Factors

Salary reduced 34%

Credit utilization 91%

EMI delayed twice

Savings reduced 42%

Recent personal loan

This builds trust and supports regulatory compliance.

Phase 5: Early Warning Detection

Instead of waiting for default.

Continuously monitor customers.

Examples

Salary not credited

↓

Alert

Balance below EMI

↓

Alert

GST filing delayed

↓

Alert

Credit score dropped

↓

Alert

Credit card utilization >90%

↓

Alert

This becomes an **Early Warning System (EWS)** rather than a static model.

Phase 6: Intelligent Decision Support

Prediction alone has little business value.

The system should recommend actions.

Example

Customer

Risk

Medium

Recommendation

Offer flexible EMI

Another

Risk

High

Recommendation

Assign Relationship Manager

Another

Risk

Very High

Recommendation

Freeze additional lending

This transforms AI into a business decision-support tool.

Phase 7: Portfolio Intelligence

Managers rarely care about one customer. They care about **the whole loan book**.

The platform should answer:

- Which branches have the highest risk?
- Which loan products are deteriorating?
- Which industries are seeing increased defaults?
- What is the projected NPA for next quarter?
- What is the expected credit loss?

Example Dashboard

Retail Portfolio

₹18,000 Cr

High Risk Customers

4.2%

Expected Default

₹310 Cr

Expected Recovery

₹125 Cr

Projected NPA

2.8%

This is executive-level information.

Phase 8: AI Credit Copilot

Generative AI can summarize customer risk for relationship managers.

Instead of reviewing dozens of fields, they receive a concise explanation.

Customer has experienced a 28% decline in salary over the past four months, has missed two EMI payments, and maintains 93% credit card utilization. Based on similar historical cases, the estimated default probability is 81%. Recommend proactive engagement and restructuring options.

This reduces manual analysis time and improves consistency.

Phase 9: Continuous Learning

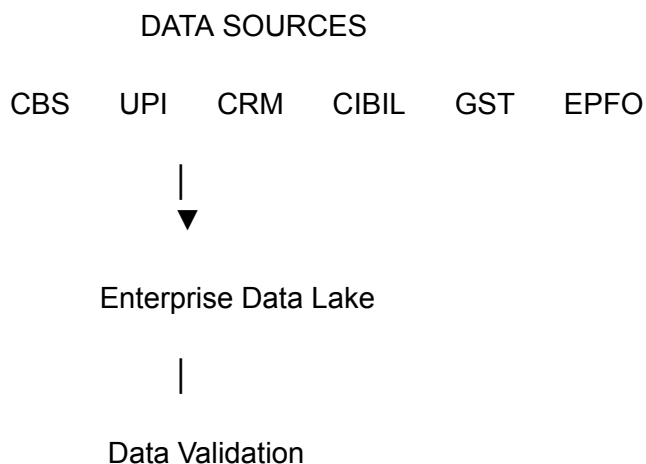
The model should not remain static.

Every month, retrain using:

- New repayments
- Fresh defaults
- Updated credit bureau data
- Economic indicators
- Recovery outcomes

This keeps predictions aligned with current borrower behavior.

Enterprise Architecture



|

Feature Engineering

|

Enterprise Feature Store

|

Explainable AI Risk Engine

|

Financial Health Score Engine

|

Early Warning Detection

|

AI Decision Recommendation

|

Portfolio Dashboard | Branch Dashboard | RM Dashboard

Future Scalability

This is where your project can shine. Instead of ending with "predict defaults," position it as a **modular banking AI platform**. The same data infrastructure and feature store can support:

Module	Business Value
Loan Default Prediction	Reduce NPAs
Credit Approval Scoring	Better lending decisions
Fraud Detection	Prevent financial losses

Cross-Sell Recommendations	Increase revenue
Customer Lifetime Value Prediction	Prioritize high-value customers
Churn Prediction	Improve customer retention
Collection Prioritization	Optimize recovery efforts
Regulatory & RBI Reporting	Faster compliance and audits

By emphasizing a reusable platform rather than a single ML model, your solution appears closer to what a bank would actually invest in: an extensible **Credit Intelligence Platform** that evolves with business needs rather than a one-off predictive model.

That framing is much more likely to resonate with judges who are thinking like banking executives rather than data scientists.

Methodology for an Explainable AI-Powered Credit Intelligence Platform

Proposed Technology Stack

Layer	Technology
Frontend	Next.js, React, Tailwind CSS, TypeScript
Backend API	FastAPI (Python)
Machine Learning	Scikit-learn, XGBoost, LightGBM
Deep Learning (Future)	TensorFlow / PyTorch
Explainable AI	SHAP, LIME
Database	PostgreSQL
Cache	Redis
Data Processing	Pandas, NumPy
Data Pipeline	Apache Airflow (Future)
Object Storage	AWS S3 / Azure Blob Storage

Authentication	JWT + OAuth2
Dashboard	Plotly Dash / Recharts
AI Copilot	OpenAI GPT / Llama 3 (RAG-enabled)
Deployment	Docker + Kubernetes
Cloud	Azure / AWS
Monitoring	Prometheus + Grafana
CI/CD	GitHub Actions

Methodology

Phase 1: Multi-Source Data Collection

The system integrates structured and semi-structured financial data from multiple sources to build a unified customer financial profile.

Internal Data Sources

- Customer Information System
- Core Banking System
- Loan Management System
- Transaction History
- Credit Card Usage
- UPI Transactions
- Mobile Banking Activity
- EMI Payment Records
- CRM Interactions
- Customer Complaints

External Data Sources (Future)

- Credit Bureau (CIBIL)
- GST
- Account Aggregator
- EPFO
- Income Tax
- RBI Economic Indicators

Output

Unified Customer Financial Profile

Phase 2: Data Engineering & Feature Store

Raw banking data undergoes preprocessing before being used for prediction.

Data Cleaning

- Handle missing values
- Remove duplicates
- Detect anomalies
- Standardize formats

Feature Engineering

Instead of feeding raw values to the model, business-specific indicators are created.

Examples

Financial Features

- Monthly Income
- Salary Growth
- Income Stability Score
- Savings Rate
- Average Account Balance

Loan Features

- EMI-to-Income Ratio
- Loan-to-Income Ratio
- Outstanding Loan %
- Remaining Tenure

Behaviour Features

- Missed EMI Count
- Average Delay in Payments
- Consecutive Missed Payments

- Credit Utilization
- Spending Volatility

Credit Features

- Credit Score
- Credit Score Trend
- Number of Active Loans
- Hard Inquiries

Time-Series Features

Past six-month trends of

- Salary
- Savings
- Account Balance
- EMI Payments
- Credit Utilization

Output

Enterprise Feature Store

This allows multiple AI models to reuse the same engineered features.

Phase 3: Data Preprocessing

The engineered dataset is prepared for machine learning.

Steps

- Label Encoding
- One-Hot Encoding
- Scaling Numerical Features
- Outlier Detection
- Feature Selection

Techniques

- StandardScaler

- MinMaxScaler
 - SMOTE (for class imbalance)
 - Correlation Analysis
 - Recursive Feature Elimination
-

Phase 4: Machine Learning Model Development

The cleaned dataset is divided into training and testing datasets.

Candidate Models

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM
- CatBoost

The best-performing model is selected based on business-centric evaluation metrics.

Evaluation Metrics

- ROC-AUC
- Precision
- Recall
- F1 Score
- PR-AUC
- Confusion Matrix

Output

Probability of Default (PD)

Example

Customer A

Default Probability

84%

Phase 5: Explainable AI Layer

Instead of producing only a prediction, Explainable AI identifies the factors contributing to customer risk.

SHAP Analysis

Displays feature contribution scores.

Example

Risk

84%

Reasons

Salary reduced

Credit utilization 92%

Three delayed EMIs

Savings declining

Outstanding debt increased

Benefits

- Transparency
- Regulatory compliance
- Increased trust among loan officers

Phase 6: Financial Health Scoring Engine

A composite Financial Health Score is calculated using multiple business indicators.

Score Components

- Credit Behaviour
- Repayment Behaviour
- Liquidity
- Debt Burden
- Income Stability
- Spending Behaviour

Example

Financial Health Score

72/100

Credit Behaviour

85

Liquidity

68

Debt Burden

55

Income Stability

80

This score enables faster business decisions beyond binary default classification.

Phase 7: Early Warning System (EWS)

The platform continuously monitors customer financial behaviour and generates alerts before defaults occur.

Example Business Rules

- Salary not credited for two consecutive months

- EMI delayed twice
- Credit utilization exceeds 90%
- Savings decrease by more than 40%
- Credit score drops significantly

Output

Early Warning Alert

Customer

ID10291

Risk Level

High

Reason

Income declining

Suggested Action

Contact Customer

Phase 8: AI Decision Support Assistant

Generative AI summarizes customer profiles into actionable insights for Relationship Managers.

Example

Customer exhibits increasing financial stress due to declining salary credits, rising credit utilization, and repeated EMI delays. Recommend proactive engagement and evaluate loan restructuring options.

This reduces manual assessment time and supports consistent decision-making.

Phase 9: Interactive Risk Dashboard

Different dashboards are provided for various stakeholders.

Relationship Manager Dashboard

- Customer Risk Score
- Financial Health Score
- SHAP Explanations
- Recommended Actions

Branch Manager Dashboard

- Branch Risk Distribution
- High-Risk Customers
- Loan Portfolio Health
- Default Trends

Executive Dashboard

- Portfolio Risk Heatmap
 - Projected NPA
 - Expected Credit Loss
 - Sector-wise Risk
 - Geographic Risk Distribution
-

Phase 10: Deployment & Continuous Learning

The solution is deployed as a cloud-native microservices platform.

Deployment

- Docker containers
- Kubernetes orchestration
- FastAPI backend
- PostgreSQL database
- Azure/AWS Cloud

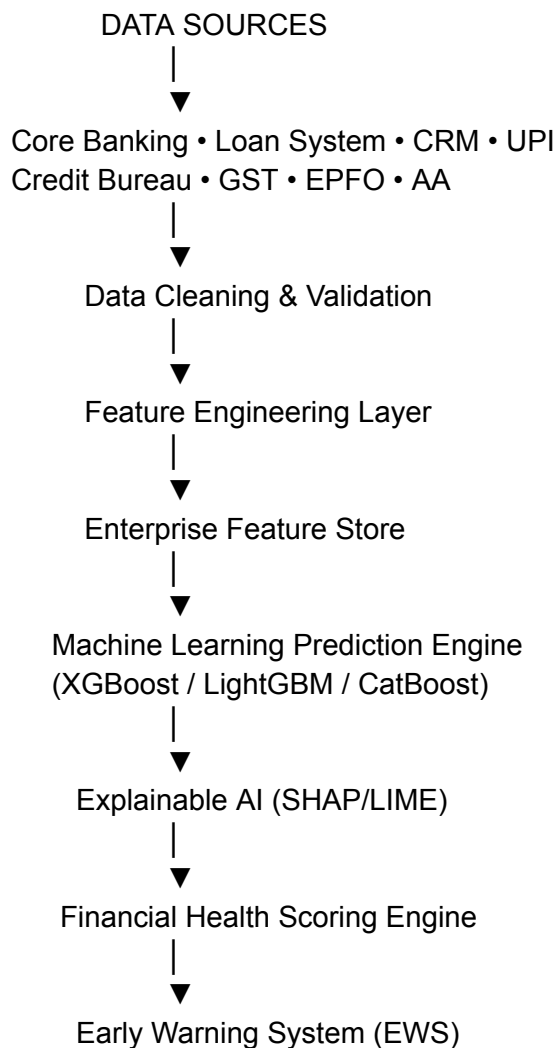
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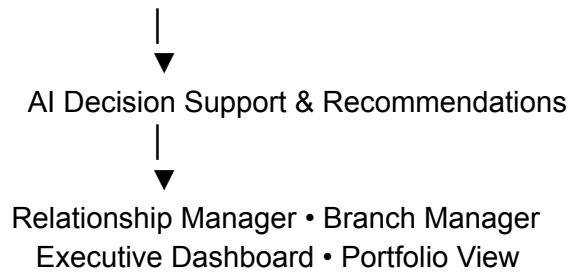
As new repayment and transaction data becomes available:

- Data is validated
- Features are updated
- Models are periodically retrained
- Performance metrics are monitored
- Prediction drift is detected

This ensures that the platform adapts to evolving customer behaviour and changing economic conditions.

Complete Workflow





Why this methodology is business-oriented

Rather than stopping at "**predicting loan defaults**," this approach delivers a scalable **Credit Intelligence Platform**. The default prediction model becomes just one component within a broader ecosystem that supports proactive risk monitoring, regulatory transparency through Explainable AI, operational decision-making for relationship managers, and portfolio-level analytics for executives. This architecture is modular, cloud-native, and extensible, allowing banks to later add capabilities such as fraud detection, loan approval optimization, customer churn prediction, and personalized financial advisory without redesigning the core platform. This is the kind of roadmap a bank could realistically adopt beyond a hackathon prototype.

I think we can take this one level higher. Instead of presenting it as:

Methodology for Default Prediction Model

Present it as:

Methodology for an Explainable AI-Powered Credit Intelligence Platform

Banks don't buy models. They buy **decision-support platforms** that reduce NPAs, improve recovery, and assist loan officers.

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Outstanding debt increased

Benefits

- Transparency
 - Regulatory compliance
 - Increased trust among loan officers
-

Phase 6: Financial Health Scoring Engine

A composite Financial Health Score is calculated using multiple business indicators.

Score Components

- Credit Behaviour
- Repayment Behaviour
- Liquidity
- Debt Burden
- Income Stability
- Spending Behaviour

Example

Financial Health Score

72/100

Credit Behaviour

85

Liquidity

68

Debt Burden

55

Income Stability

80

This score enables faster business decisions beyond binary default classification.

Phase 7: Early Warning System (EWS)

The platform continuously monitors customer financial behaviour and generates alerts before defaults occur.

Example Business Rules

- Salary not credited for two consecutive months
- EMI delayed twice
- Credit utilization exceeds 90%
- Savings decrease by more than 40%
- Credit score drops significantly

Output

Early Warning Alert

Customer

ID10291

Risk Level

High

Reason

Income declining

Suggested Action

Contact Customer

Phase 8: AI Decision Support Assistant

Generative AI summarizes customer profiles into actionable insights for Relationship Managers.

Example

Customer exhibits increasing financial stress due to declining salary credits, rising credit utilization, and repeated EMI delays. Recommend proactive engagement and evaluate loan restructuring options.

This reduces manual assessment time and supports consistent decision-making.

Phase 9: Interactive Risk Dashboard

Different dashboards are provided for various stakeholders.

Relationship Manager Dashboard

- Customer Risk Score
- Financial Health Score
- SHAP Explanations
- Recommended Actions

Branch Manager Dashboard

- Branch Risk Distribution
- High-Risk Customers
- Loan Portfolio Health
- Default Trends

Executive Dashboard

- Portfolio Risk Heatmap
 - Projected NPA
 - Expected Credit Loss
 - Sector-wise Risk
 - Geographic Risk Distribution
-

Phase 10: Deployment & Continuous Learning

The solution is deployed as a cloud-native microservices platform.

Deployment

- Docker containers
- Kubernetes orchestration
- FastAPI backend
- PostgreSQL database
- Azure/AWS Cloud

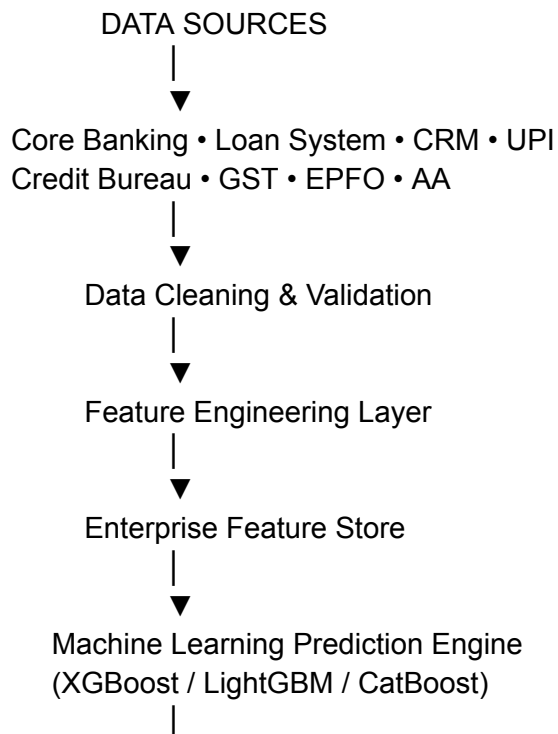
Continuous Learning

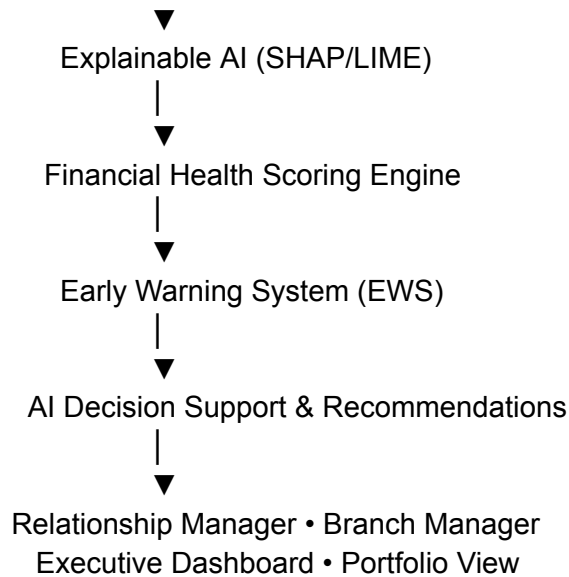
As new repayment and transaction data becomes available:

- Data is validated
- Features are updated
- Models are periodically retrained
- Performance metrics are monitored
- Prediction drift is detected

This ensures that the platform adapts to evolving customer behaviour and changing economic conditions.

Complete Workflow





Why this methodology is business-oriented

Rather than stopping at "**predicting loan defaults**," this approach delivers a scalable **Credit Intelligence Platform**. The default prediction model becomes just one component within a broader ecosystem that supports proactive risk monitoring, regulatory transparency through Explainable AI, operational decision-making for relationship managers, and portfolio-level analytics for executives. This architecture is modular, cloud-native, and extensible, allowing banks to later add capabilities such as fraud detection, loan approval optimization, customer churn prediction, and personalized financial advisory without redesigning the core platform. This is the kind of roadmap a bank could realistically adopt beyond a hackathon prototype.